# REP 105 SPATIAL TRANSFORM TREE & COVARIANCE PROPAGATION

Decoupling Continuous Dead Reckoning from Discontinuous Global Localization Across Embodied Coordinate Frames

## COORDINATE FRAME KINEMATIC HIERARCHY

SE(3) RIGID TRANSFORMS & DRIFT

### `earth` Earth-Centered Geodetic Frame (ECEF / UTM / WGS-84)

GLOBAL PLANETARY

Static planetary origin for GNSS / RTK constellations and geospatial coordinates

`T_earth_to_map (Fixed Datum)`

### `map` World-Fixed Metric Map Reference Frame

GLOBAL CONSISTENT

**Characteristics:** Non-drifting global frame. Long-term accuracy bounded by map survey.

**Update Properties:** Low rate (1–10 Hz); **Discontinuous step-jumps allowed** on relocalization.

T_map_to_odom(t) [Discontinuous Step Jumps on Loop Closure]

### `odom` Locally Smooth Odometry Reference Frame

STRICTLY CONTINUOUS

**Characteristics:** Dead reckoning (wheels + IMU + VO). Continuous & differentiable ($C^1/C^2$).

**Drift Behavior:** High rate (50–200 Hz); **Continuous drift: Σ_drift(t) ∝ Q_rw · √t** without step jumps.

T_odom_to_body(t) [High-Rate Continuous Kinematic Pose]

### `base_link` Robot Physical Center / Chassis Origin (body)

PHYSICAL CHASSIS

**Kinematic Anchor:** Center of mass / drive axle midpoint. Swept volume origin.

**Consumers:** Motor torque & velocity controllers consume odom→base_link directly.

| `camera_optical` | `lidar_link` | `imu_link / radar` |
| --- | --- | --- |
| Z-forward / X-right / Y-down | Laser emitter / range center | Specific force & Doppler array |
| `T_body_cam [Σ_ext]` | `T_body_lidar [Σ_ext]` | `T_body_imu [Σ_ext]` |

`z_obs ∈ F_sensor` **Physical Observation with Measurement Covariance Σ_z ∈ $\mathbb{R}^{3\times3}$**

Full Spatial Mapping: **p_map = T_map^odom · T_odom^body · T_body^sensor · z_obs**

## 1. THE DOWNSTREAM CONSUMER SPLIT (REP 105 INVARIANT)

### A. Real-Time Tracking & Enforcement (50 Hz – 1 kHz)

**Consumes:** `T_odom_to_body(t)` (Smooth Odometric Frame)

**Systems Invariant:** Must be **strictly $C^1/C^2$ continuous**. Any step jump creates infinite derivative jerk ($\dddot{x} \to \infty$), tripping motor inverter overcurrent.

### B. Global Path Planning & Memory (1 Hz – 10 Hz)

**Consumes:** `T_map_to_body = T_map^odom · T_odom^body` (Global Frame)

**Systems Invariant:** Absorbs SLAM loop closures & GPS corrections in `T_map^odom` via discrete jumps without shaking the tracking controller.

## 2. SPATIAL UNCERTAINTY COMPOSITION ACROSS FRAMES

**First-Order Spatial Covariance Chain:**

`Σ_map ≈ J_odom·Σ_drift·J_odomᵀ + R_body·Σ_ext·R_bodyᵀ + R_tot·Σ_z·R_totᵀ`

- **Extrinsic Mounting Drift (Σ_ext):** Thermal baseline expansion
  `Δb = α · L · ΔT (e.g., 25°C rise on 120mm Al bar = 0.069mm shift)`
- **Long-Range Lever-Arm Error:** Angular mount drift $\theta$ yields lateral error $\delta x \approx z \cdot \theta$ (0.29° error at 10m range = **50.0 mm lateral bias**).

## 3. MANDATORY OBSERVATION HANDOFF CONTRACT

- **Capture Timestamp $t_0$ (Midpoint):** Eliminates velocity lag $\Delta x = v \cdot \tau$.
- **Reference Frame Key F_sensor:** Binds measurement to kinematic tree.
- **Uncertainty Envelope Σ_z:** Weights Kalman innovation; avoids chatter.

**SAFETY RULE: Downstream enforcers reject any packet missing $t_0$, F_sensor, or Σ_z**